

# LostLink: Cross-Organisation Lost & Found Matching

## CS5224 CLOUD COMPUTING - TEAM 16 PROJECT PROPOSAL

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## 1. Motivation & Business Case

### 1.1 Problem Statement

Lost-property management is fragmented across individual organisations. A person may lose an item while moving between a university, public transport, a shopping mall, hotel, office, or event venue, yet not know where the loss occurred. The usual process is to search each organisation separately or repeatedly check a found-item list. For example, NUS asks users to check its Found Items Public List and return later if the item has not yet appeared. Meanwhile, staff must manage reports, search their organisation's inventory, respond to enquiries, and verify ownership claims manually. This creates repeated effort for users and administrative workload for organizations.

### 1.2 Project Idea

LostLink is a privacy-aware, cross-organisation Lost & Found matching service. A user submits one lost-item report with description, approximate location and time, and optionally photos and the places travelled through. Participating organizations keep their found-item inventories private, but authorise LostLink to search them for potential matches. New lost and found reports are compared using image, text, spatial and temporal similarity. When a strong match is detected, the user is notified that a potential item is held by a partner organization; sensitive item details remain hidden until ownership is verified by the organisation.

### 1.3 Cloud Service Provider

LostLink will be developed and deployed using Amazon Web Services (AWS) as the cloud service provider. AWS services selected for the prototype include Amazon Cognito for authentication, Amazon API Gateway for API access, AWS Lambda for application logic, Amazon S3 for image storage, Amazon DynamoDB for application data, Amazon SQS for asynchronous processing, Amazon Bedrock for AI-based matching, and Amazon SES for email notifications.

## 1.4 Competitive positioning

Existing platforms validate the need for digital Lost & Found, but they mainly operate either as public/community listings or as isolated organisational systems. LostLink focuses on private **cross-organisation matching**.

| Service                                                       | Current Approach                                                                                          | Cross-organization private matching | Key distinction/limitation                                                              |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------|
| <a href="https://lostandfound.io">https://lostandfound.io</a> | Free community lost/found listings; matching reports are surfaced; hotels can publish selected held items | No                                  | Community/public discovery; <b>independent</b> organisations decide what becomes public |
| Nuvlost                                                       | AI semantic + visual matching, automated notifications, multi-location organizational dashboards          | No                                  | Data is not shared across organisations                                                 |
| LostLink (our)                                                | One user report searches authorised inventories of multiple participating organizations                   | Yes - proposed core feature         | Cross-organisation matching while inventories remain private until verification         |

## 1.5 Business Model

LostLink will be free for individuals. Users will not pay to report a lost item, receive alerts, or submit a claim. Revenue will instead follow a B2B SaaS model. Organisations pay for operational features such as inventory management, staff access, matching & claim tracking. Pricing could be tiered by locations or item volume, for example, a small single-location plan, a multi-location organisational plan, and custom enterprise pricing. Advanced features such as analytics and audit history, configurable data-retention policies, API integration, SSO, and custom branding are potential extensions for future development. The value to organisations is reduced staff effort in manually answering enquiries, searching records, and coordinating returns. An initial rollout could target organisations within a shared campus or neighbourhood, building a useful local network before expanding to additional locations.

## 2. Use Case & Scope

### Individuals can but not limited to:

- Register, verify email, sign in, recover account access.
- Submit, edit or withdraw a missing-item report, with optional photos and visited places.
- Receive notifications, review or dismiss similar-item suggestions for their report and receive candidate/claim updates.
- Submit ownership evidence, answer staff questions and track claim decisions.
- View collection instructions after approval and see the recorded handover status.

### Organisation staff can but not limited to:

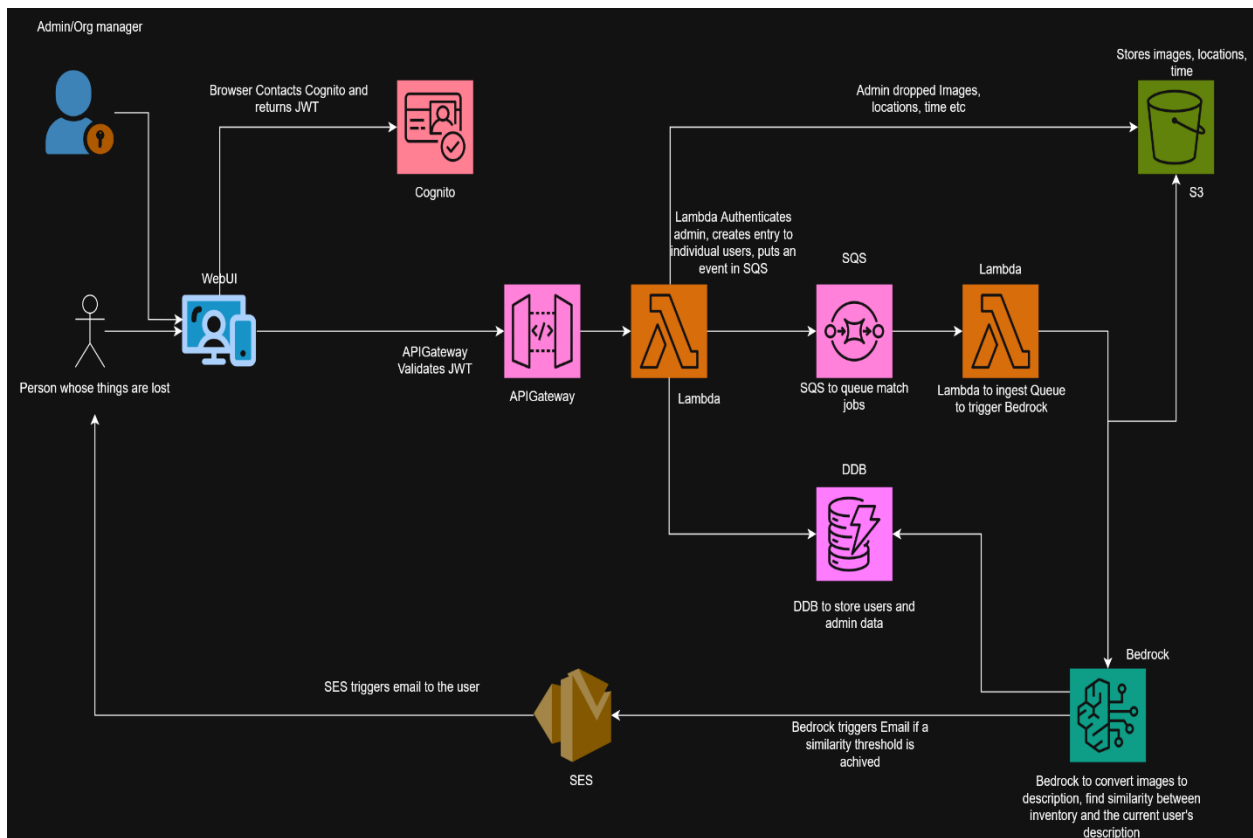
- Sign in to their assigned organisation's workspace.
- Register found items and search or update their organisation's inventory.

- Review claim evidence, request more information, and approve or reject claims.
- Reserve approved items, record physical handover and close related cases.
- Withdraw unavailable items and update affected claimants.

### Prototype Scope:

The prototype will provide account management, lost-item reporting with optional photos and visited places, staff inventory management, automated matching and notifications, ownership-evidence submission, staff review, claim-status tracking and handover recording. Reports and item availability can be updated or withdrawn. The demonstration will use simulated organisations with manually entered inventories. We assume that participating organisations authorise LostLink to process their item data for matching and that staff remain responsible for ownership decisions and physical handover. Real institutional integrations, SSO, SMS, shipping and billing are outside the prototype scope.

## 3. Architecture



## Authentication & User Interaction

The **Web UI** provides the frontend for both users and administrators. **Amazon Cognito** manages user authentication and access control, while **API Gateway** acts as the communication layer between the frontend and backend. Authenticated requests are passed through API Gateway to **AWS Lambda**. Lambda will enforce permissions using the authenticated user identity, staff role and server-controlled organisation membership. Individuals will access their own reports and claims, while staff will access their organisation's inventory and related claims. **DynamoDB** will store report metadata, organisation identifiers, claims and match results. **Amazon S3** is used alongside this flow to store uploaded images.

## Processing & Notification

New or updated lost-item and found-item reports will create matching jobs in **SQS**. A worker **Lambda** will retrieve the relevant records from **DynamoDB** and images from **S3**, invoke **Bedrock** when embeddings need to be generated, and calculate candidate scores. It will store matching results in **DynamoDB** and call **SES** to notify users when a candidate exceeds the notification threshold. Notifications will be tracked by report-item pair to limit repeated alerts during reprocessing. This asynchronous workflow allows report submission to complete without waiting for matching and notification.

## AI-Based Matching

**Amazon Bedrock** provides the AI capabilities of the system. It can process information from lost and found items, such as descriptions and images, to identify potential similarities and generate matching results. The AI processing is triggered through the asynchronous processing pipeline and can use the item information stored in **S3 and DynamoDB**. The resulting match information is then passed back to the application for storage and, where appropriate, notification through **SES**.

# 4. Implementation and Evaluation Plan

## 4.1 Implementation Plan

| Phase             | Plan                                                                                                                                                                                          |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Foundation        | Explore components, tradeoff, Set up AWS resources, Cognito user pools, API Gateway endpoints, Lambda functions, DynamoDB tables and S3 buckets. Establish role and organisation identifiers. |
| Core Workflows    | Implement user registration, lost-item reporting, image upload, found-item registration, organisation inventory views and case/status management.                                             |
| Matching Pipeline | Add SQS job creation and Lambda consumption; integrate Bedrock multimodal inference; implement candidate scoring, thresholding and match persistence.                                         |

|                          |                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Claims and notifications | Implement candidate notifications through SES, ownership evidence submission, staff review, approval/rejection, reservation and handover recording.                  |
| Testing and evaluation   | Populate ground-truth data, run matching experiments, measure system performance/cost, conduct user evaluation and refine threshold/configuration based on findings. |

## 4.2 Evaluation Plan

**Matching quality.** We plan to evaluate approximately 100 lost-item reports against 200 found-item records across simulated organisations. For the initial prototype, ground-truth data will be produced using a combination of synthetic and real examples. LLMs can generate varied lost/found descriptions, times, locations, and distractor reports, while image-generation tools can create visually related item images. A smaller set of everyday physical objects will also be photographed from different angles and backgrounds.

Metrics include: Hit@1, Recall@5, Mean Reciprocal Rank (MRR). Experiments will compare text-only, image-only, image&text, image&text&location&time variants. This will show whether contextual information genuinely improves retrieval rather than merely increasing system complexity.

**Notification quality.** Service metrics will use a fixed observation window and unique report-item alerts, such as: alert precision, no-match false alert rate, missed-notification rate, alert burden (incorrect alerts/report), notification latency.

**User and staff evaluation.** Participants complete tasks using separated organisational lists vs. LostLink. Tasks cover reporting, cross-organisation enquiry, no-match cases, evidence submission, staff approval/rejection. Measures can include task completion, errors, assistance requests, time to complete task, etc.

**System evaluation.** Experiments will vary the number of stored items, concurrent submission. Metrics will include matching latency, throughput, queue-processing time, and estimated cloud cost.

## 5. Conclusion & Future Work

LostLink proposes a privacy-aware cross-organisation lost-property network rather than another isolated lost-and-found database. Its main contribution is allowing a user to report an item once while the system continuously searches authorised inventories across multiple organisations using image, text, location, time, and potentially journey information.

The prototype will focus on cross-organisation matching, privacy-aware notifications, claim verification, and a manageable cloud architecture suitable for the project timeframe. Future work includes real integrations with universities, malls, hotels, and transport operators; multilingual reports; continuous model improvement from confirmed matches and fraud-resistant ownership verification; expanding the scope to individuals to store lost items and automatic verification using our AI service to remove the manager or organization based verification.